

Research Interests

System design for wireless communication and sensing, mmWave system design, Reconfigurable metasurface, Integrated sensing and communication, RF systems for Quantum applications, RF design for Space applications

Education

2020–present **Indian Institute of Science.**

MTech(Research) and PhD in Electrical Communication Engineering, **CGPA: 9/10**

Thesis: Design of a Multiport Antenna and an Optimized Metasurface for the demonstration of Wireless Channel Modulation

Advisor: Prof. K. J. Vinoy, Expected completion: November 2026

2015 **Indian Institute of Engineering Science and Technology, Shibpur.**

B.E. in Electronics and Telecommunication Engineering, **CGPA: 8.15/10 (First Class with Honours)**

Collaborative Research Experience

May-July, **Dept. of EE, The Pennsylvania State University, University Park, USA.**

2025 Visiting Researcher on the project 'Sensing and Communications with RIS' with Prof. Ram Narayanan, worked on AI-assisted channel prediction for metasurface-enabled media-based modulation.

Scholarships and Awards

- Received the **Best student paper award (second place)** in IEEE MAPCON 2025.
- Recipient of the **Prime Minister's Research Fellowship (PMRF)** for Doctoral studies.
- Finalist in the **student best paper award competition** in IEEE MAPCON 2022.
- Recipient of the **Indian Oil Merit Scholarship** for Undergraduate Studies.
- Recipient of the **Merit-cum-Means Scholarship** from the State Govt. of West Bengal for Undergraduate Studies.

Research Skills

- Worked on Antenna, metasurface, and Passive circuit design with CST Microwave Studio, Altair Feko and Keysight ADS.
- Worked on USRP, PlutoSDR, and GNU Radio for radar and wireless communication.
- Designed the VNA interface for radiation pattern measurement in the anechoic chamber.
- Worked on Particle swarm optimization for Metamaterial design.
- Worked on Python and MATLAB for signal processing for wireless communication.

Journal Articles and Patents

- K. J. Vinoy et al., "A Digitally Reconfigurable Metasurface for Communication in Scatter-Rich Multipath Channel," **Indian Patent** 435646, issued June 26, 2023.
- S. Roy, K. J. Vinoy, "A Compact Monopulse Antenna With Dual-CP Sum Patterns for Applications in Internet of Drones," accepted for publication in **Transactions on Vehicular Technology**.
- S. Roy, K. J. Vinoy, A. Chockalingam, "A Multi-Port Receiver for Decoding A Channel-Modulated Radio Frequency (RF) Signal and Method Thereof," **Indian Patent application** in process.

- S. Roy, K. J. Vinoy, A. Chockalingam, "A Noncoherent Channel Modulation Scheme Exploiting Orthogonal Radiation Patterns in a Two-Port Receiving Antenna," Manuscript in preparation.

Conference Proceedings

- S. Roy, A. Roy, K. J. Vinoy, "Implementation of Noncoherent Media-Based Modulation With a Reconfigurable Metasurface," in Proc. IEEE Microw. Antennas Propag. Conf. (MAPCON), Dec. 2022.
- S. Roy et al., "A mmWave Phased Array for Joint Radar and Communication," in Proc. IEEE Microw. Antennas Propag. Conf. (MAPCON), Dec. 2023.
- S. Roy, K. J. Vinoy, "Integrated Design of a Compact, Multiport Antenna and Feed Network With 2-D Monopulse Patterns," in Proc. Asia-Pacific Microwave Conference (APMC), Nov. 2024.
- S. Roy, K. J. Vinoy, "Binary Particle Swarm Optimization for Design of a Reconfigurable Transmissive Unit Cell," in Proc. IEEE Microw. Antennas Propag. Conf. (MAPCON), Dec. 2024.
- S. Roy, K. J. Vinoy, "A Wideband Dual-Feed Microstrip Antenna With Pattern Diversity," in Proc. IEEE Microw. Antennas Propag. Conf. (MAPCON), Dec. 2024.
- S. Roy, K. J. Vinoy, "Pattern-Diversity Receiving Antenna for a Channel Modulation Scheme," in Proc. IEEE Int. Symp. Antennas Propag. North Amer. Radio Sci. Meeting (AP-S/CNC-USNC-URSI), Jul. 2025.
- S. Roy, Z. Fu, K. J. Vinoy, R. M. Narayanan, "Constellation Prediction for Reconfigurable Metasurface-Inspired Media-Based Modulation Using Residual Neural Networks," in Proc. IEEE Microw. Antennas Propag. Conf. (MAPCON), Dec. 2025.

Teaching Experience and Public Outreach

Demo	Demonstrated 'Angle of Arrival Estimation With mmWave Quasi-monopulse Radar' for the general public in the IISc Open Day 2023.
PMRF TA	Offered an open online course through ISSS on 'Microstrip Antennas and Applications' in 2023.
Demo	Demonstrated 'Visualization of the function of the different blocks in a transceiver chain with Matlab and over-the-air data transmission using PlutoSDR' for the general public in the IISc Open Day 2024.
IISc TA	Performed teaching assistantship for the course - 'Radio Frequency Integrated Circuits and Systems' by Prof. K. J. Vinoy, ECE Dept. IISc in 2024.
PMRF TA	Offered an open online course through ISSS on 'Design and Simulation of RF Circuits' in 2024.
Demo	Demonstrated 'Smart Home Enabled by Radar' for the general public in the IISc Open Day 2025.
PMRF TA	Offered an open online course through ISSS on 'Array and Metasurface for Wireless Communication' in 2025.

Relevant Course Work

Graduate	Antenna Theory and Practice, Radio Frequency Integrated Circuits and Systems, Computational Electromagnetics, Digital Signal Processing, Design of Cyber-Physical Systems, Mathematics for Electrical Engineers, Electromagnetic Metamaterials, Stochastic Models and Applications, Advanced Electromagnetics, Communication System Design
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Work Experience

Designation	Junior Executive Officer - Air Traffic Controller (2015 - 2019).
Company	Airports Authority of India , NSCBI Airport, Kolkata, India.

Activities

- Founding Member of the IEEE MTT/AP-S Student Branch Chapter, IISc.
- Previous Member of the ECE Dept. Wellness Committee, IISc.
- Participated in Basic Life Support training in IISc.
- Participated in **Science for Rural India Group, IISc** activities.
- Undergraduate industrial training completed from Siemens India, on Elevator Designing Using Programmable Logic Controller.

Language

- Bengali, English, Hindi.

Interests

- Table Tennis, Badminton, Running, Cooking, Football.

Referees

- Prof. K. J. Vinoy
Professor
Department of Electrical Communication Engineering
Indian Institute of Science, Bengaluru, India
email: kjvinoy@iisc.ac.in
- Prof. Ram M. Narayanan
Professor
Department of Electrical Engineering
The Pennsylvania State University, University Park, USA
email: rmn12@psu.edu
- Prof. Debdeep Sarkar
Assistant Professor
Department of Electrical Communication Engineering
Indian Institute of Science, Bengaluru, India
email: debdeep@iisc.ac.in